

7.11 PreQuiz: Logarithmic and Exponential Functions — Markscheme

1. [Maximum mark: 6]

Lina invests \$2400 at 6 % per annum, compounded quarterly.

(a) Quarterly rate = $\frac{0.06}{4} = 0.015$, so the quarterly growth factor is 1.015. **A1**

Accept 1.5 % with a feedback note that, for a growth “factor”, 1.015 is the preferred form.

(b) There are $5 \times 4 = 20$ quarters. **M1**

$V = 2400(1.015)^{20} = \$3232.45$ **A1**

(c) Need the smallest integer n with $2400(1.015)^{4n} > 3400$. **M1**

$4n > \frac{\ln(3400/2400)}{\ln(1.015)}$, so $n > 5.85 \dots$ **A1**

Smallest full-year $n = 6$ years. **A1**

Alternative method (table): show both adjacent rows $2400(1.015)^{20} = 3232.45$ (at $n = 5$) and $2400(1.015)^{24} = 3430.84$ (at $n = 6$), then state $n = 6$. A single-row table earns full marks but should receive a feedback note recommending the adjacent row to justify the crossover.

2. [Maximum mark: 6]

$$M(t) = 5000e^{kt} \text{ and } M(2) = 7200.$$

(a) $5000e^{2k} = 7200$, so $e^{2k} = 1.44$. **M1**

$$k = \frac{\ln 1.44}{2} = 0.1823\dots$$
A1

(b) $M(6) = 5000e^{6k}$, using their value of k . **M1**

$$M(6) = 14\,929.92\dots, \text{ so } 14\,930 \text{ members.}$$
A1

Accept 14 901 (obtained by substituting rounded $k = 0.182$) with a feedback note: do not round intermediate values; carry full precision through and round only at the end.

(c) Set $5000e^{kt} = 10\,000$, so $e^{kt} = 2$. **M1**

$$t = \frac{\ln 2}{k} = 3.80 \text{ years.}$$
A1

Accept 3.81 (obtained by substituting rounded $k = 0.182$) with the same feedback note about not rounding intermediate values.